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ORIGINAL ARTICLE

Impacts of climate change on electrical subsidies' public policies: the case of Mexico

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Abstract
Several countries around the world subsidize electrical tariffs. Mexico's residential tariff structure is among the most complex and subsidized, depending on blocks of consumption and local temperature. In 2016 the residential electricity subsidy was 100 billion Mexican Pesos. To understand the increases in residential electricity consumption at the municipal level in Mexico in a climate change context, we used a machine learning approach based on gradient boosted trees to test how electricity consumption is related to population, GDP, temperature, and precipitation, while controlling for month of the year and for the country's different states. We evaluated the impact of consumption increases, coupled with the future behavior of the current tariff policy, on the future increase of the total subsidy in real terms. The subsidy would grow to 132.4 billion MXN in real terms by 2030, and 203.6 billion MXN in 2070 under a business as usual scenario; climate change makes the subsidy rise by 2.2% by 2030 and 5.3% by 2070. Projected increases in temperature should be incorporated explicitly in the planning of the National Electric System, as the subsidy would continue to grow in real terms. This situation is worsened by the current tariff structure based on the average temperature of the locality. A tariff change proposal is discussed; it maintains the current structure while ameliorating the distributional incidence, with a tariff increase of 18% for the lowest income decile and 80% for the highest income bracket, and reducing future subsidy increases under climate change by 50% by 2070.

Keywords Subsidies · Climate change · Climate impacts · Demand forecasting · Tariff policy · Mexico

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1 Introduction

Electrical tariffs are subsidized in many countries around the world (Komives et al. 2005). Increasing Block Tariffs (IBT) are the most common subsidy and consist of differentiated prices for each additional block of electric consumption. IBT schemes are generally highly regressive since lower-income deciles receive a higher percentage of their total bill, but higher-income deciles receive a higher total subsidy (Komives et al. 2005).

Another form of subsidy is a spatially differentiated tariff according to climate, in the understanding that, in tropical areas, hotter climates are prone to higher consumption, while the inhabitants of such regions may not have a similar or higher disposable income in comparison to regions with climates leading to lower consumption. In a warming world with increasing energy demands (Auffhammer and Mansur 2014; Botzen et al. 2020), a regressive electricity tariff structure may further vulnerate the financial situation of both public utilities and low income users.

Several efforts have been made to reform electric tariffs and reduce subsidies (CONECC 2018). Turkey introduced a single tariff above the total cost of electricity which resulted in a large price hike. Colombia offers a subsidy only below a subsistence level of consumption but is plagued by cross-subsidies among different regions. India, Iran, and Indonesia transitioned to Direct Benefits Transfer mechanisms, coupled with a long communication campaign.

Mexico's tariff structure is among the most complex in the world due to the subsidy structure (Komives et al. 2009). Most of the total subsidy is not directed to lower deciles, mainly because of climate-differentiated tariffs (Komives et al. 2009). The residential tariff in Mexico (tariff 1) is heavily subsidized with an IBT system (CONECC 2018). The first 75 kWh consumed monthly are charged (January 2025) at 1.063 Mexican Pesos (MXN) (basic step); the next 65 kWh at 1.290 MXN (intermediate step); and the remaining consumption at 3.777 MXN (surplus step). The first two steps are located most of the time below the total cost of electricity.

The rest of the residential tariffs (1 A, 1B, 1 C, 1D, 1E, 1 F) are climate-subsidized. In each increasing tariff letter, the consumption limit of each step is higher, and the charge per kWh is cheaper. The thresholds for each tariff block are set to maximum monthly averages of 25, 28, 30, 31, 32, and 33 °C to enter tariffs 1 A to 1 F, respectively. The price for any of these tariffs is the same in all of Mexico, generating regional cross-subsidies. The DAC (*Doméstica de Alto Consumo*, High Consumption Residencial Tariff) tariff is a punitive rate for users who exceed an annual consumption threshold, which for tariff 1 is 3,000 kWh. This tariff is five times more expensive and consists of a fixed charge and a flat rate for all the kWh consumed. It is the most expensive of all tariffs in Mexico, and one of the highest in the world (Aburto 2007).

The temperature tariffs have emerged in a disorganized manner, and when the DAC tariff was created in 2002 to alleviate the financial burden on the public national electric utility company (CFE) and to reduce the regressivity of the tariff structure, there was strong social pressure that triggered the creation of 1E and 1 F tariffs, thus increasing the total subsidy (Aburto 2007). Most of the users that moved from DAC to 1E and 1 F were higher-income, air-conditioning users. In 2017, a major tariff reform was implemented concurrent with the liberalization of the electricity market; however, residential tariffs were left unchanged.

The combination of rising incomes, a complex and regressive tariff structure differentiated by climate, and an increasing financial strain of government entities in Mexico form a perfect storm for multiple interlocking crises due to impacts of climate change in energy consumption. The electric system development program (PRODESEN) considers economic growth, population growth, climate seasonality, fuel prices, the price of electricity, and electricity losses as drivers for electricity demand growth (SENER 2020). Despite the omission in the national plan, an increase in electricity consumption is expected under climate change scenarios (Botzen et al. 2020). Cooling degree days (CDD) are generally calculated by adding, for each day, the difference between the mean temperature and an arbitrary base value, which represents the outdoor temperature beyond which a building would theoretically need to be cooled to keep the occupants comfortable; they are also expected to increase (Corrales-Suastegui et al. 2021). Tejeda-Martínez et al. (2022) also studied the increase in electrical consumption for major Mexican cities due to increased cooling demand, estimating it as a function of the change in human hygrothermal comfort. Hygrothermal comfort is achieved through the control of temperature and relative humidity to obtain a state of balance of the human body with respect to sensible and latent heat fluxes.

In this study, we implemented a machine learning model – which can be better suited for forecasting highly nonlinear, correlated variables than traditional econometric models (Friedman 2001; Chen 2021) – to evaluate current and future residential electricity consumption in Mexico and how it may change in the future, considering the interrelated effects of climate change and socioeconomic development. We generate future consumption projections at the municipal level, improving the spatial resolution and spatial extent of previous studies. The subsidy would grow from 100 billion MXN in 2016 to 132.4 billion MXN in real terms by 2030, and 203.6 billion MXN in 2070 under a business as usual scenario. We then evaluate the current distributional incidence of the subsidy, and how it may vary in the future under climate change conditions. Finally, we propose a tariff change that would reduce the total subsidy and ameliorate the distributional incidence, while retaining the basic tariff structure to avoid widespread opposition to the new policy. The moderate tariff change results in a tariff increase of 18% for the lowest income decile and 80% for the highest income bracket and reducing future subsidy increases under climate change conditions by 50% by 2070.

1.1 Data and methodology

1.1.1 *Climatological variables*

We use 2010 to 2016 as the current study period (this period was chosen due to data availability issues explained in Sect. 2.2), 2030 as the near future, 2050 as an intermediate future, and 2070 as the distant future. Current monthly climate data over the period 2010–2016 were used as predictors in the statistical model. The reference climatology was calculated for the period 1995–2014 following the recommendations of the Intergovernmental Panel on Climate Change (IPCC 2021).

We used the CHIRTS dataset (Funk et al. 2019) to represent temperature for the current period temperature and for calculating the Cooling Degree Days (CDD). In this study, we captured the variation in human comfort standards in different climates using a spatially variable base temperature for CDDs, so that the temperature at which people need air-conditioning depends on the locality to which they are acclimatized to. This base temperature was selected as the climatological annual mean temperature (1995–2014) for each grid point. The indoor comfort temperature in places without mechanical ventilation is a linear function of the mean outdoor temperature (ASHRAE 2010). Moreover, Mexico has a low air conditioning penetration level (16% in 2016) (IEA 2018), so the mean outdoor temperature can be used to evaluate people's comfort preferences. Building insulation can significantly alter energy consumption even within the same climate, but in Mexico around 95% of residential buildings still don't have any kind of insulation (Rosas-Flores and Rosas-Flores 2020), so it is not considered a relevant factor nation-wide.

The CHIRPS database (Funk et al. 2015) was used to represent precipitation, as a proxy for relative humidity because both variables are usually strongly correlated (Wilby and Wigley 2000), and the future projection data used only has precipitation. A high relative humidity, combined with high temperatures, implies higher demand for air-conditioning systems, as happens in Mexico's coastal regions, which for the most part have tropical climates and high annual precipitations due to the influence of hurricanes and easterly waves. We calculated a variable that is equal to the accumulated precipitation on days in which the temperature is higher than the climatological average.

Each climate variable was spatially averaged to obtain a single value per municipality in order to match socioeconomic data's spatial resolution. Averaging climate data at municipality scale may introduce a potential bias due to the uneven distribution of population centers, but there are no future GDP and population projection databases with a higher spatial resolution for the period of study.

1.2 Socioeconomic variables

We consider the number of users and total electricity consumption for all types of residential consumers and users per municipality (CFE 2018). We chose to use CFE billing data instead of income survey data (ENIGH) because the latter's sample is not representative of different tariff zones; however, subsidy conclusions between both datasets are largely comparable (Komives et al. 2009) We obtained the sum of all tariffs and selected a single tariff per municipality, the one corresponding to most of the consumption. Consumption data is only available from 2010 to 2017, while high-resolution temperature data is available until 2016. Therefore, we limited the study period to 2010–2016 to match the coverage of the different datasets.

Within a specific municipality, we corrected for data values that were either abnormally low or large. A similar problem is reported by Komives et al. (2009). Accordingly, the lower threshold was set to 30 kWh. The excluded municipalities represent 4.57% of the total number, but only 0.01% of the total population and 0.13% of the total electrical consumption. The excluded values introduce a large bias to the model and makes the validation runs fail, and the selected threshold is also where the results of the model start to stabilize. The average number of users per municipality during the study period is 13,270 with a maximum value of 527,935. The average consumption per user/month/municipality is 89 kWh with a maximum value of 906 kWh. The annual information was interpolated into monthly values from the monthly consumption data by state (SENER 2021). We calculate each user's electricity bill using the full formulae for CFE's complex IBT, climate-dependent tariff structure, including the cutoff value for entering the non-subsidized tariff (CFE n.d.).

We used the annual gross domestic product per state (GDP) in millions of constant 2003 Mexican pesos (MXN) (INEGI 2015) and to downscale the data to the municipality level for it to be consisted with the scale of the rest of the data, we used the methodology established previously for an analogous problem in Haro et al. (2021). This methodology consists in converting the income component of the municipal human development index (INAFED 2013) into per capita income and scaling it according to INEGI state GDP information. We also used population per municipality from the 2010 census and the 2015 intercensal survey (INAFED 2013); the rest of the years have been linearly interpolated.

For the model, we also analyzed the inclusion of variables such as the price of fossil fuels, and the type of tariff in each municipality as a proxy for the price of electricity, factors that have been found the relevant for the United States (Branch 1993). However, for the case of Mexican residential electricity consumption for the period 2010–2016, none of these variables proved to have a relevant effect on consumption and the number of users.

To simplify the calculation of the electricity payment, we averaged the consumption between 2010 and 2016 and calculated it with 2013 prices, available on CFE's website (CFE n.d.). The residential electricity subsidy in 2016 was 100 billion MXN (CONECC 2018), which corresponds to 90.7 billion MXN in 2013 prices, using Mexico's Central Bank inflation data (Banxico 2021).

It is difficult to estimate the real cost of CFE's electricity delivered to the end-user. The total cost is equal to this marginal cost plus the costs of transmission, distribution, and CFE's variable and financial operating costs. However, this information is not publicly available, which prevents the direct calculation of the total subsidy. The subsidy and cost of electricity are related by:

$$\text{Subsidy} = \text{Total cost} - \text{Sale price}$$

In this article, we use CFE's full methodology to calculate the sale price, but use an indirect approach to obtain the total cost using the value of the DAC tariff:

$$\text{Total Subsidy} = \text{a } \Sigma \text{ N}_\text{i} \text{ x DAC Tariff Price}_\text{i} - \text{Sales}$$

where Total Subsidy is the yearly sum of subsidy given to all residential users, DAC Tariff Price_i is the flat rate portion of the DAC Tariff for each of the i-th administrative CFE regions, N_i is the number of users in region i, Sales is the yearly sum of the total sale price for all users calculated according to the current tariff structure, and α is a scaling factor chosen such that the left and the right-hand side of the equation are equal. The spatial differentiation of the DAC Tariff allows the consideration of different levels of subsidy according to the local cost of generation. The scaling factor is akin to the profit margin, so this indirect method assumes that it is homogeneous throughout the country. The factor α was found to be 0.66. Making CFE's costs publicly available would allow more accurate estimates.

The climate projections were generated from the emulator AIRCC-CLIM (Estrada et al. 2022). The climate change scenarios were calculated using the delta method for constructing fields of change (IPCC 1994). This bias-correction method consists in taking the differences between the historical climatology from a numerical climate model and a climatology of future climate conditions obtained from the same model. The result is added to an observed dataset of higher resolution. For precipitation, the percent change is calculated.

Currently, AIRCC-CLIM cannot project extreme events such as heat waves that may also be relevant for residential electricity consumption, so we have not included these factors.

Temperature and precipitation changes were calculated by a stochastic experiment of 1,000 realizations emulating 39 different Global Circulation Models (GCM) using the AIRCC-CLIM emulator. The ensemble averages of the 1,000 realizations were used for projecting the impacts of climate change on electrical subsidies.

1.3 Future projections

We chose two sets of combinations of the Representative Concentration Pathways (RCP) and of the Shared Socioeconomic Pathways (SSP): the RCP8.5-SSP5 which is a high-emissions, high-socioeconomic development scenario, and the RCP4.5-SSP2 which represents more moderate trajectories. The RCPs represent the radiative forcing in W/m^2 that would be achieved by 2100 (IPCC 2021). RCP8.5 corresponds to a 4–6 °C warming by 2100, and the RCP4.5 to a 2–3 °C increase in global mean temperatures. The Shared Socio-economic Pathway SSP5 corresponds to a “Fossil-fueled World” with high population growth, strong economic development, and significant international cooperation and trade. In contrast, the SSP2 is a “Middle of the Road” scenario, where social, economic, and technological trends do not shift markedly from historical patterns (Riahi et al. 2017).

Future economic and population growth data for the chosen SSPs were taken from the integrated CLIMRISK model, with a resolution of 0.5°x0.5° (Estrada and Botzen 2021). We have used CLIMRISK’s module for disaggregating socioeconomic data projections to the grid level, without considering climate damages to the economy, since the complexity of the feedbacks exceed the scope of this research, where the main point is to isolate the effect of temperature increase on energy consumption. An increased energy consumption is expected to affect disposable incomes and thus dampen economic growth; the same effect may be expected of other climate impacts. This results in a lower GDP growth, that is most commonly analyzed in the literature as a GDP loss as compared with a scenario without climate loss. However, in a scenario with climate change, GDP is still expected to continue increasing, albeit at a lower rate. Care must be taken not to assume that the resulting GDP growth is due to climate change, but rather despite it, and that coupling a macroeconomic model with our energy consumption model in further research may help elucidate these feedbacks.

Having socioeconomic data with relatively low resolution in comparison with the typical municipality size (there are municipalities even smaller than the climate data 0.1°x0.1° grid) can be an important source of uncertainty, although the socioeconomic scenarios are intrinsically uncertain. With the historical data from SSP5 and the future data socioeconomic and climate data, we calculate the percentage change per municipality and multiply it by the current data (Fig S3).

1.4 Statistical model

Residential electricity consumption in the United States has been linked to income, the price of electricity and fossil fuels, the use of appliances, the size of the household, and climate (Branch 1993). For statistical models of residential electricity consumption in Latin America, quantile regression econometric models have been used (Hancevic and Navajas 2015). Panel data methods are optimal for studying the elasticity of consumption to climate since they allow compensating for the effect of unobserved variables (Auffhammer and Mansur 2014).

The use of machine learning methods for problems that are traditionally addressed with econometric techniques is incipient. Machine learning provides new tools for economic research and is useful for a different set of problems than econometric methods, whose main motivation is parameter estimation for a deep analysis of the drivers of a phenomenon (Mullainathan & Spiess, 2017; Charpentier et al. 2018). When the main motivation is forecasting, machine learning methods can outperform traditional econometric methods such as panel data (Chen 2021) since they are particularly useful for highly nonlinear data and sets of strongly correlated variables (Friedman 2001). Here we used a machine learning model to forecast residential electricity consumption in Mexico. Another motivation against using a panel data model is that there are much more municipalities than time steps and controlling for each municipality would imply the loss of too many degrees of freedom.

We use a *Gradient Boosted Trees* machine learning model, specifically *XGBoost* (Chen and Guestrin 2016) and *Sklearn* (Pedregosa et al. 2011); it is a useful, efficient algorithm for nonlinear data that greatly reduces the probability of overfitting. Gradient Boosted Trees is a machine learning algorithm with combining several methodologies to make for a more robust algorithm (Friedman 2001). A decision tree evaluates an aspect of some predictor and makes a binary decision. A random Forest corrects decision trees' overfitting problem by training a large number of decision trees that only have partial access to a subset of the system's predictors and calculating an average for continuous predictands. The last step consists of Gradient Boosting, where the decision trees are trained in succession, and in each training round more weight is given to the previous error, thus reducing the total error with great efficiency without sacrificing the predictive capacity with new data.

Transforming the data with a logarithm worsened the performance of the model, so it was left untransformed. The regression hyperparameters were fitted from a range of options: *learning rate* between 0.1 and 0.5, *min child weight* between 0 and 25, *min split loss* between 0 and 1.2, *alpha* between 0 and 12, and *lambda* between 0 and 12. The total sample was randomly divided into 70% for training, and 30% for validation. We selected the subset of variables and hyperparameters that minimize the training criterion, which in this case was the *root mean squared error* (RMSE).

Two models were created, both with the same hyperparameters: *learning rate* = 0.1, *min child weight* = 25, *min split loss* = 0.3, *alpha* = 0, *lambda* = 0, *colsample by tree* = 0.3, *max depth* = 50, *number of estimators* = 100. The chosen model for the number of users per municipality was:

$$\text{Users} \sim (\text{Pop},\ \text{GDP},\ \text{PCI}, \text{Year},\ \text{State})$$

where Users is the number of electric users per municipality each year, Pop is the population per municipality each year, GDP is the gross domestic product per municipality each year, PCI is the per capita income per municipality each year, and State is a set of dummy variables for each of the 32 states. Different possible combinations of variables were tested, selecting the combination where variables were significant and where the RMSE was lower and the R^2 was higher. Even though the model uses highly correlated variables (Pop, GDP, PCI), the nonlinear nature of the gradient-boosted tree machine learning algorithm is not affected by such issues. This set of variables resulted in an RMSE of 507.3 and an R^2 of 0.99983 for the training set.

The other model predicts the electrical consumption per user:

$$\mathrm{Cons_{us}} \sim (\mathrm{T_{means}},\ \mathrm{CDD},\ \mathrm{Pre_{Tmeans}},\ \mathrm{Dens_{pop}},\ \mathrm{PCI},\ \mathrm{Month},\ \mathrm{State}) \tag{1}$$

Where $\mathrm{Cons_{us}}$ is the consumption per user in a given municipality in a given month of each year, $\mathrm{T_{mean}}$ is the monthly mean temperature in a given municipality, CDD is the cooling degree days, calculated using a base temperature equal to the mean climatological temperature, $\mathrm{Pre_{Tmean}}$ is the monthly accumulated precipitation on days with a temperature equal or higher than $\mathrm{T_{mean}}$, and $\mathrm{Dens_{pop}}$ is the population density in a given municipality in a given month and year. The model produces an RMSE of 6.410 and an R^2 of 0.9850 for the training set.

The model was validated by a cross-validation partitioning strategy where 100 random different validation subsets of the data were obtained (the size of each validation set is 30% of the total data) and the RMSE and R^2 was calculated for each of them (Ghojogh and Crowley 2019). The range of RMSE for the 100 sets was 533.3-621.3 for the Users model and 10.877–11.816 for the Consumption per User model and the R^2 had a range of 0.99976–0.99982 and 0.9510–0.9575 respectively (Fig S2). The obtained values compare favorably to the actual training set, suggesting an adequate forecasting power without overfitting.

We tested the same predictor variables using a simple linear regression instead of a machine learning algorithm to analyze whether our choice of a more complex method was valid. The Users model has an RMSE of 6048.4 and an R^2 of 0.97669. While the R^2 value is high, the RMSE is an order of magnitude higher than with the gradient boosted trees model. The linear model for the Consumption per User resulted in an RMSE of 31.03 and an R^2 of 0.66259. In this case the RMSE is 5 times larger and the R^2 indicates is low. This demonstrates that the use of a more complex, nonlinear model is necessary to correctly forecast this residential electricity consumption data. The correlation matrices (tables S3 and S4) also show correlation coefficients higher than 0.5 for several variables. Since most econometric models are based on linear regression, highly correlated variables and nonlinear behavior make machine learning models such as gradient boosted trees a much better fit for this study.

Some municipalities with a very low initial number of users show either very small and unreasonable values of users per person or very large ones. Therefore, results were cropped to be between 0.11 and 1.6 users per person, which is consistent with Komives et al. (2009). The climate change scenario for 2030 had 9.4% of municipalities corrected, which corresponds to just 1.11% of the population, and 1.16% of total consumption.

Additionally, we averaged all the variables for each of the 32 states. The model was trained again with the same hyperparameters, and the results were comparable to the model trained with data at the municipality level (Table S2). Although a few municipalities are much smaller than the socioeconomic data grid, this does not result in an appreciable bias. The following analysis makes use of the municipalities model The model projects the electric consumption per user, after which we calculate each user's electricity bill. Afterwards, the model also projects the number of users per municipality, with which we calculate the total payment. Using the methodology explained Sect. 2.2, we derive the future subsidy projection from this value.

2 Results

Despite being regressive, subsidies in Mexico are less unequal than the income distribution, with a quasi-Gini coefficient of 0.16 and a Gini coefficient of 0.49 respectively, the latter calculated with information from GCIP (Global Consumption and Income Project) (Jayadev et al. 2018). While the smallest value of the income distribution is 0, for a subsidy it can go as low as -1, exceeding the line of perfect equality.

Fig.1.a shows the spatial distribution of tariffs per municipality for the period 2010-2030 according to CFE information, which corresponds to the actual distribution. Since the regions with tariff 1 accumulate many small municipalities, tariff 1 is the most common. Also, the largest population and wealth centers in the country are concentrated in those municipalities, so they correspond to most of the subsidies. In general, a transition of warmer tariffs towards the coasts can be seen due to their humid climate, although the most subsidized tariffs are concentrated in the north of the country, except for a handful of municipalities in the south. The highest concentration of high-temperature tariffs is in the northwest, although per capita income in the region is several times higher than in the south of Mexico, which has lower subsidized tariffs.

Fig.1.b, shows the difference between the tariff distribution according to the temperature calculation methodology established by CFE and the actual distribution, where positive numbers (colored red in the graph) indicate that the real tariff is higher than the calculated one. There is a significant number of municipalities that are misclassified and benefit from a higher subsidy than they should, particularly in the north of the country and in the southeast. This is especially notable in municipalities that should have a low tariff but are adjacent to others with a very high tariff. This is consistent with what was mentioned in this article's introduction and in the historical review of tariffs by Aburto (2007).

Fig.2.a shows how the type of tariff would change across Mexico between 2010 and 2030 in the RCP8.5-SSP5 scenario. There is a strong loss of municipalities in tariff 1, with some even jumping to tariff 1C and a strong increase in the most subsidized tariffs throughout the country. Notably, several municipalities would have a jump of more than one tariff, with the average change in tariff per municipality being +0.421, where a +1 change is a jump between adjacent tariffs, like 1A and 1B. Fig. 2.b shows the number of municipalities in each type of tariff in the current period (2010-2016) and the future under the RCP8.5-

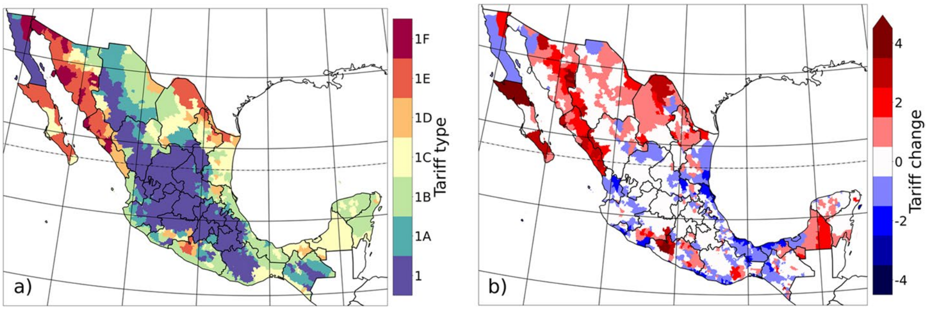

Fig. 1 Residential tariff type across Mexico: **a**) real tariff for the current period (2010-2016); **b**) difference between real tariffs and the theoretical ones for the current period(2010-2016), where a red value indicates that the real tariff is higher than the theoritical value.

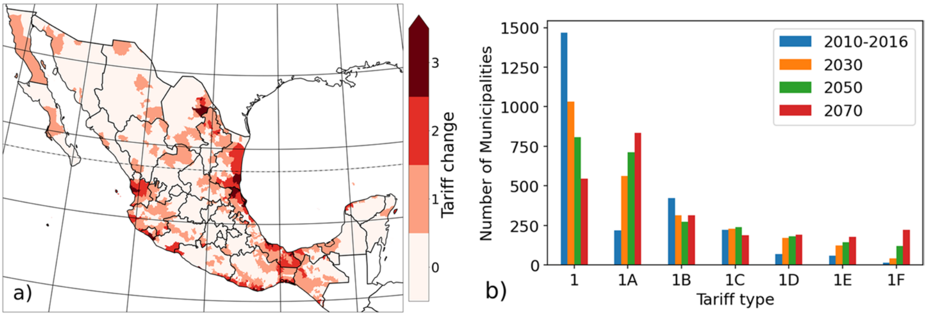

Fig. 2 **a**) Tariff change between 2030 and 2010 in the RCP8.5-SSPS scenario, **b**) municipalities in each tariff type in the current period (2010-2016) and projected values for the RCP8.5-SSP5 scenario.

SSP5 scenario. There is a transition towards higher temperature tariffs, which also have a higher subsidy. Tariffs 1, 1B, and 1C show a decrease over time, while the rest increase. There is a considerable increase in tariff 1A between the current period and 2030, and in tariff 1F between 2030 and 2050 and between 2050 and 2070. By 2070, the tariff with most municipalities would be 1A, while tariff 1F would have a larger number of municipalities than 1C, 1D, or 1E.

Fig.3.a shows the projected increase in consumption between 2010 and 2030 in the RCP8.5-SSP5 scenario. The national increase in consumption is 59%, which corresponds to 2.34% per year. The model projects a total increase in the number of users of 52.6%, which would most probably be related to the decrease in the average number of individuals per family associated with the fertility decrease that Mexico is expected to undergo in the following decades.

Fig.3.b shows the difference between consumption in 2030 considering the effects of climate change and without taking them into account. These estimates provide an indicator of the temperature effect of RCP8.5 given the change in population and GDP of SSP5. The difference is 8.1% more consumption attributable to climate change. By 2070 the difference grows to 18.8%, which is larger than the 12% increase by 2100 calculated by Botzen et. al (2020), although they consider the reduction of heating demand with fossil fuels. Several regions stand out for having a strong regionally homogeneous increase.

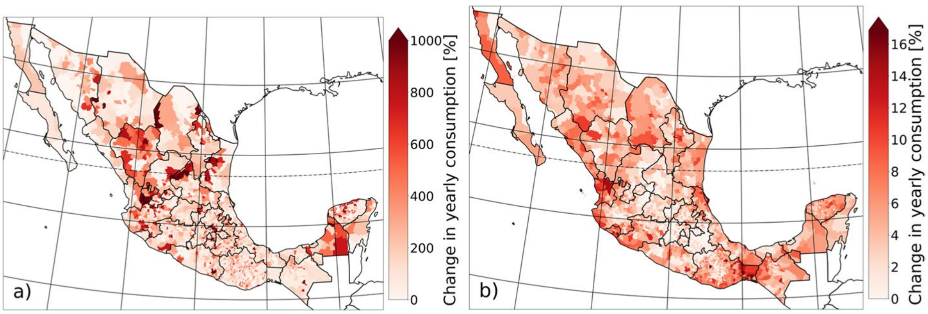

Fig. 3 Change in total consumption per municipality: **a**) between 2010 and 2030 in the RCP8.5-SSP5 scenario, where values larger than 1,000% are saturated; **b**) difference between the climate change scenario (RCP8.5-SSP5) and the constant temperature one (only SSP5), where larger values indicate that consuption is higher when the temperature is considered and values than 17% are saturated.

The future subsidy structure would remain regressive, with a quasi-Gini coefficient of 0.08 in 2030, 0.03 in 2050, and 0.03 in 2070 under the RCP8.5-SSP5 scenario (Fig S1.b). The regressivity of the subsidy would decrease slightly due to many high-consumption users getting into the surplus tier. Fig. 4.a shows the current and future distributional incidence of the subsidy. All deciles show an increase in subsidy from 2010 to 2070, with the smallest increase from 2050 to 2070. The increase is larger in the middle part of the income brackets, which helps to reduce the quasi-Gini coefficient, albeit with a smaller benefit to the lowest income deciles. From 2030 onwards the last decile would receive a higher subsidy than the ninth decile.

Fig.4.b shows the future evolution of the subsidy in real terms. The subsidy would increase over the study period, although the growth rate decreases by the end of the century due to users entering the surplus tier. The subsidy grows from a yearly average of 90.7 billion MXN in 2010-2016 to 132.4 billion MXN in 2030 in the RCP8.5-SSP5 scenario.

There are two main factors which can increase the total subsidy. If a municipality changes to a higher-temperature tariff rate, keeping the same consumption per user, the subsidy increases, and if the tariff rate remains unchanged, but the consumption per user increases without reaching the DAC tariff, the subsidy also increases. In this case, most of this increase in the total subsidy is due to the increase in consumption per user in each municipality (due to climate change and socioeconomic factors), with the transition to higher-temperature tariff rates playing a small role (exclusively due to climate change); this is made evident by comparing the increase in subsidy keeping the tariff types unchanged and by allowing them to change. If the tariff types were to remain constant, in 2030 there would be a 130.6 billion MXN subsidy in 2013 prices (2.4 billion MXN less than by allowing tariffs to change).

While the effect of the tariff change is smaller, it is evident even today, where the subsidy in real terms with the theoretical tariffs would be 82.5 billion MXN. The difference in yearly subsidy in 2030 between the climate change scenario (RCP8.5-SSP5) and the constant temperature scenario (only socioeconomic change, SSP5) would be 2.2%, while in 2070 the difference grows to 5.3%. In 2070, the total subsidy would amount to 219.3 billion MXN in the RCP-SSP5 scenario.

The model was also run using a milder socioeconomic and climate change scenario, RCP4.5-SSP2, in order to evaluate the sensitivity of the model to these factors. This scenario, known as the "Middle of the road" scenario, considers that historical trends con-

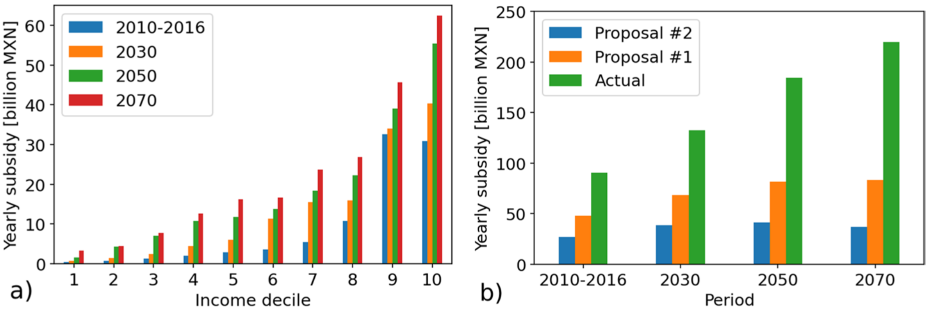

Fig. 4 **a**) Distributional incidence of the subsidy in the RCP8.5-SSP5 scenario. **b**) Current period (2010-2016) and projected subsidy evolution for the present tariff structure and two proposals in the RCP8.5-SSP5 scenario.

tinue into the future and thus has a lower population and GDP growth, while temperature increases stay slightly below 3°C by 2100. In this case, the average municipality temperature change was 0.7 °C and a 3.7% increase in precipitation between 2010 and 2030. The national population increases by 11.7%, and GDP by 65.7% in the same period. This results in a consumption increase of 50.7% and 48.7% more users in 2030, and a total subsidy of MXN 126.7 billion. The average tariff change is +0.36 per municipality. This represents a 6.9% consumption increase against a scenario without climate change and a 1.7% increase in the total subsidy. As can be seen, the difference in 2030 between scenarios RCP4.5 and RCP8.5 is small, highlighting the existence of short-term impacts even in the face of increased climate ambition. However, for 2070, the RCP4.5 scenario results in an average consumption increase of 155.6% and a total subsidy of MXN 203.6 billion. This represents a 10.2% consumption increase against a scenario without climate change (only SSP2) and a 3.8% increase in the total subsidy. The consumption increase exclusively due to climate change is 18.8% in the high temperature scenario (SSP5-8.5) and only 10.2% in the mild temperature scenario (SSP2-4.5). This factor, coupled with a lower population growth, results in a difference of MXN 15.4 billion in subsidies by 2070 between both scenarios. In this temporal horizon, the consumption and total subsidy increase diverges between both scenarios, indicating that timely climate action could avoid the worst impacts. Supplementary Fig. S2 also evaluates different sensitivities to the exogenous variables and shows that results are consistent.

Additionally, we make a tariff restructuring proposal to evaluate public policy measures that could mitigate the increase in the total subsidy. The proposed change must be simple to try to avoid the negative effects that happened in 2002 with the introduction of the DAC, 1E, and 1F tariffs. We also avoid proposing changes in the DAC tariff, such as lowering the threshold or increasing the rate because high-income users tended in recent years to buy photovoltaic systems of a size just big enough to jump back to the subsidized tariff (ABM 2017; Chacon-Anaya 2016), eroding the reduction in the financial burden for CFE. Therefore, the objectives of the proposed tariff change are reducing the impact of climate change on subsidies, moderating the transfer of economic burden to the users, reducing the regressivity of the subsidy, and avoiding the potential pitfalls of past tariff reforms. At the same time, it shows that a large portion of the tariff issues can be solved without substantially changing the tariff structure, which is something that has not been explored before in other reform proposals, as mentioned in the discussion. Our proposal consists in keeping the DAC tariff as it currently stands, including the different thresholds needed to jump from 1A-F tariffs to DAC. Additionally, we propose the elimination of all the intermediate steps in all tariffs, including intermediate, intermediate low, and intermediate high. The summer threshold between the intermediate high and the high steps in the 1F tariff is the same as the yearly average threshold to change to DAC tariff and is one of the main reasons that make this tariff the most subsidized of all, even at very high levels of consumption. Under our proposal 1, all charges in every step are raised by 10%. This proposal is named"Proposal 1" hereafter. A second, stricter, proposal ("Proposal 2"), drives down the threshold of each consumption step (Table S1).

Both proposals result in the accomplishment of the objectives fixed. The subsidy would be greatly reduced in the present to 48.3 and 26.9 billion MXN with Proposals 1 and 2, respectively, and the future growth of the subsidy would be slower, reaching 68.7 and 38.5 billion MXN in 2030 under the RCP8.5-SSP5 climate change scenario (Fig. 4.b). The regressivity of the subsidy is reduced, and Proposal 2 even results in a negative quasi-Gini coefficient of –0.2, indicating a progressive subsidy (Fig S1.c). The total yearly subsidy goes down by more than 50% in Proposal 1 and by more than 70% by 2070 (Fig. 5.a) in the RCP8.5-SSP5 scenario. However, it must not be forgotten that this relief of CFE's finances would come at the cost of an increased electricity expenditure per user. For the lowest income decile, this results in an 18% hike in Proposal 1 and a 30% hike in Proposal 2. The tariff raise would be much higher for the highest income bracket, 80% in Proposal 1 and 120% in Proposal 2, although the difference between both proposals is considerably smaller, as can be observed in Fig. 5.a. The increase in payments is also smaller for tariff 1: 29% and 73%, than for tariff 1F: 99%, and 127% (Fig. 5.b). While the increases in tariff 1 are large, most of it would go towards air-conditioning, high-income users.

Finally, a sensitivity analysis has been conducted for the machine learning model. The model was trained 100 times with random training subsets of the data; future projections were generated for the 2030 horizon with the RCP8.5-SSP5 scenario. The minimum and maximum values for the total subsidy with the current tariff were 129.5 and 132.5 billion MXN, for the subsidy with Proposal 1 67.5 and 69.3 billion MXN, and for the Proposal 2 the values were 37.9 and 39.2 billion MXN, respectively. The reported results fall within the range of the uncertainty given by the model training, and the range of results leads to similar conclusions.

3 Discussion

There are likely inaccuracies in the first and last deciles in the Lorenz curves since we're taking the spatial average for a municipality that may have a diversity of socioeconomic levels, especially in urban areas. The aggregated data indicates that no municipality is in DAC tariff, so the number of users paying a tariff substantially above marginal cost is zero. However, during the 2010-2016 period, there were on average 1.4% of users in DAC tariff, corresponding to 4.8% of total residential consumption. No municipality appears in the high intermediate consumption tier, although many individual users are. This would considerably increase the total subsidy in the higher temperature tariffs (1C-F). These non-linear effects may offset each other, but it is difficult to estimate this with aggregated data. Additionally,

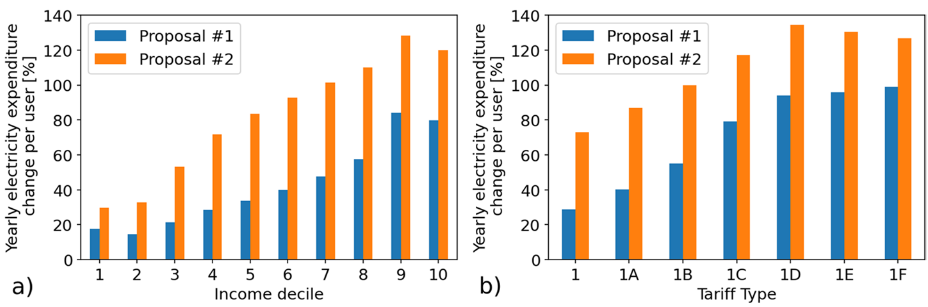

Fig. 5 **a**) Distributional incidence of the tariff increase for each proposal for the 2010-2016 (current) period. **b**) Tariff increase distribution across different tariff types for each proposal for the 2010-2016(current) period. Both proposals eliminate the intermediate consumption steps and raise the charges by 10%. Proposal 2 additionally reduces the thresholds for each consumption step.

aggregated data eliminates local inequalities in consumption and income. The national Gini coefficient calculated with the PCI data per municipality is 0.34 (Fig S1.a). However, we consider that the information presented here is representative at a large scale since it agrees with other studies on the distribution of the electricity subsidy in Mexico such as (Komives et al. 2009) and (CONECC 2018); additionally, more than 80% of the municipalities in Mexico have low and low-medium inequality (Gallardo del Ángel et al., 2021), a fact which suggests that consumption per users are relatively homogenous in a given municipality, as has been assumed given the granularity of the data.

There are interrelations between socioeconomic and climate factors that have not been considered in this paper. Although the central zone of the country has a preponderance of electricity tariffs with lower subsidies (Fig. 1.a), the redensification of Mexico's large cities under the SSP5 scenario would increase the urban heat island effect (UHI), which could cause entire cities of several million inhabitants to switch to a more subsidized tariff. Tejeda-Martinez et al. (2022) consider the UHI, but since it depends on the total population and urban morphology, they do so only for the largest cities in the country. However, UHI happens in any human settlement, even in rural areas. A correlation between population density and UHI could be found to evaluate it across the whole country, given the average urban morphology in urban and rural areas in Mexico. This is not a trivial matter and could be the subject of a future study.

Additionally, there is a complex interrelation between air conditioning adoption, rising incomes (which have been slower than expected), technological change, and electricity prices (Davis and Gertler 2025). A model that considers feedbacks in behavioral responses to electricity prices, models air conditioning adoption and future electricity prices might be much better at capturing the system dynamics surrounding cooling. However, among these factors, the most uncertain may be future energy prices which depend on decarbonization, energy mixes, and technological change. Future directions may be the coupling of the model presented in this paper to an integrated assessment model such as GCAM that can model future electricity prices under different climate change scenarios (Wilkerson et al., 2015).

The apparent widespread incidence of municipalities being in a lower tariff than they really ought to be taken with care. The CFE database indicates multiple tariffs for some municipalities, suggesting that the tariff type is set at the locality level. The information aggregated per municipality does not allow us to elucidate these details, and the difference in tariff type might not exist for some municipalities. However, the figures presented (1.a and 1.b) indicate that this is a generalized issue in large regions of the country and that it is also present in municipalities that are part of a large urban area covering several municipalities.

PRODESEN 2020-2035 projects a total national electricity consumption growth of 3.0% from 2021 to 2026 and 2.8% from 2020 to 2035, which is slightly higher than the increase obtained here for the RCP8.5-SSP5. PRODESEN's scenario doesn't include climate change, although it considers a widespread use of electric vehicles. This effect is difficult to resolve spatially across the country on a municipal scale, since the municipal evolution of electromobility adoption would depend on the car share, per capita income, recharging infrastructure implementation, electric utilities' policies, government policy, smart grid implementation, among other factors that are too uncertain to project into the future; we have thus not included electromobility evolution in our model. This consumption increase would only further increase the future subsidy, so we consider the conclusions to still be valid. PRODESEN's projected growth without considering electric vehicles is 2.51%, which is slightly larger, but still comparable to the value we report in this paper, especially since our main focus is the comparison between a climate change scenario (RCP8.5-SSP5) and one without climate change (just SSP5). Additionally, the short-term PRODESEN scenario considers an acceleration of the economy after the COVID-19 pandemic, while our model doesn't consider this effect. PRODESEN does not consider the increased consumption from industrial and residential electrification, and we did not try to consider it since the spatially distributed evolution has the same complexities as the ones found for electromobility.

The increase in consumption by 2030 in the RCP8.5-SSP5 scenario would be large enough to bring many municipalities into the surplus consumption tier when previously there were none. This is likely to result in a large share of users on the DAC tariff when disaggregating consumption; this should not be seen as a relief to CFE's finances. The 2002 tariff change demonstrates that there is a great risk that this will trigger social pressure for municipalities with more users on the DAC tariff to increase to a higher temperature tariff level (Aburto 2007). On the other hand, the training data of the model does not have any municipality in DAC tariff, let alone any that switched to or from it, and the price of electricity has not been explicitly considered, which prevents the elasticity of consumption from being established. This prevents modeling the expected drop in electricity consumption of users who see their payment multiplied several times when they enter the DAC tariff.

The difference in yearly subsidy in 2030 between the climate change scenario (RCP8.5-SSP5) and the constant temperature scenario (only SSP5) indicates that the effect of climate change is subordinate to socioeconomic effects. However, the effect of temperature on total consumption is much larger, and the effect on subsidies is masked by the widespread change of users to the surplus, non-subsidized step. This could lead to social pressure to increase the level of subsidy.

For the tariff restructuration proposals, both CFE's and the citizen's financial situation must be taken into consideration. If the main objective is reducing the inequality of the subsidy and alleviating CFE's financial burdens, the lowest-cost-for-the-consumer option is Proposal 1. If the objective is to reduce as much as possible the subsidy, Proposal 2 could be the choice, although additional measures should be put in effect to avoid social unrest due to the steep hike in utilities and the considerable impact on the lower socioeconomic groups.

Previous tariff proposals include reducing the DAC threshold (CONECC 2018), the negative effects of which have been discussed above. Komives et al. (2009) warn against the small gains and large negative effects of such a change and that even small reductions in subsidies would result in a significant burden for the lowest income deciles; our proposal reduces this burden on the lowest income households. Aburto (2007) proposes gradually increasing the intermediate step charge, which is in the same spirit as the proposals made here. During the 2013 energy reform in Mexico, there was a proposal to include a new residential tariff based on local marginal prices, which was never actually implemented.

The consideration of the social marginal cost of pollution in addition to marginal generation costs in the pricing of electricity tariffs around the world has been discussed (Borenstein & Bushnell 2022). Hancevic et al. (2019; 2022) have discussed the effect that IBT tariffs have on suppressing the incentives for energy efficiency. Any efficiency measure reduces the subsidy for CFE and is desirable for the company but the marginal effect for consumers is small to nonexistent; therefore, reducing the regressivity of the tariff structure also increases the incentives for energy efficiency measures. These authors made a tariff change proposal for Mexico but argued that the marginal cost proposal has negative distributional incidence implications and would be politically unacceptable. Therefore, they combine it with a means-tested Direct Benefits Transfer mechanism. Whether a marginal cost system could be introduced without social unrest and whether the inequality correcting mechanism would serve its purpose in a corruption-ridden country is not clear. Other proposals include the compensation of the social and environmental cost of electricity by redirecting the subsidy to renewable energy investment, energy efficiency, or health care (CONECC 2018); the savings from the proposals discussed here could also be directed to these means.

4 Conclusions

In this study, we have used a machine learning model to generate projections of residential electricity consumption in a future scenario considering socioeconomic factors and climate change. Most studies of the kind use traditional econometric methods such as panel data. This paper has used a not yet widely applied machine learning model, but which is highly flexible in handling different variables (e.g., climate and socioeconomic data) and allows for an easy implementation of sensitivity analysis. Currently, a significant number of municipalities are on a higher subsidy tariff than they should according to their average temperature, which increases the total subsidy from 88.5 to 90.7 billion 2013 MXN for the period 2010-2016. In the future, many municipalities would move out of tariff 1, and by 2070, tariff 1A would be the most common in the country.

The increase in consumption between 2010 and 2030 would be 59%, while the difference between consumption in a scenario with and without climate change is 8.1%. This emphasizes the importance of considering the effects of climate change in the planning of electricity generation and consumption to increase national resilience and adaptation to climate change.

The residential electricity subsidy in Mexico is regressive with a quasi-Gini coefficient of 0.16. In the future, subsidy inequality would be reduced to a coefficient of 0.03 in 2070, although the tenth income decile would significantly increase its total subsidy. The subsidy would grow to 132.4 billion MXN in real terms by 2030 in the RCP8.5-SSP5 scenario. Temperature increase due to climate change makes the subsidy rise by 2.2% by 2030 and 5.3% by 2070.

Climate change will help worsen the state electric utility's (CFE) precarious financial situation by increasing the total amount of residential subsidies and its impacts may be evident as soon as 2030. Even in a more moderate climate change scenario, such as the RCP4.5- SSP2, subsidies increase and have visible effects by 2030. However, impacts are much larger by 2070 in the RCP8.5, highlighting the value of timely climate action.

Changes in the tariff structure need to be subtle and gradual since a radical change could be counterproductive, as happened in 2002 with the introduction of the DAC, 1E, and 1F tariffs. Two tariff change proposals were presented; the first reduces the inequality of the subsidy, cuts the electric subsidy by 50% by 2070, and implies a moderate increase of 18% for the lowest income decile, compared with an 80% increase for the highest income bracket. The second proposal makes the subsidy progressive and cuts the electric subsidy by 70% by 2070 but implies a large hike of 30% for the lowest-income decile. An additional effect of reducing the regressivity of an IBT tariff is to promote energy efficiency measures, which can in turn reduce the total subsidy by reducing energy demand.

Tariff structure changes are not the only policy interventions that might ameliorate the financial situation of CFE. Other measures not analyzed in this paper might include energy efficiency measures, whether it be through regulation of appliances or by programs that exchange old appliances for new ones, time-of-use rates to better align rates to generation costs, and even novel proposals that have been made such as installing residential photovoltaic systems to displace the subsidized generation. This is especially important not only due to increased demand caused by climate change, but also due to the imminent increase in demand due to electrification caused by the energy transition.

Future work will focus on how the proposed tariff could be gradually introduced, on the inclusion of climate change damages and energy efficiency feedbacks to the model, on better estimations for increases in demand due to electrification, and in developing better climate emulators that allow to include indicators for projections of extreme weather events that influence residential electricity consumption. The results presented here suggest concrete measures that policymakers can take to increase the resilience of electricity networks, utilities, and consumers in the face of climate change:

a) Consumption and demand forecasts must include the effects of climate change, where multiple scenarios should be studied to include the uncertainty in the future evolution of climate in the planning process.
b) Climate impacts could be stronger by the end of the current decade, so immediate action is of utmost importance.
c) A lower emissions scenario results in lower impacts, stressing the importance of timely action and the potential to pay for mitigation strategies with the avoided burdens of a higher-emissions scenario.
d) Current policies may have unexpected consequences and behaviors in the face of climate change, so it is important to study their evolution to avoid unexpected pitfalls.
e) A residential tariff reform is urgently needed in Mexico. The new tariff structure will need to be carefully designed to avoid the issues presented in past reform efforts.
f) It is possible to reduce the regressivity of the subsidy and reduce the total amount of subsidy without significant changes to the current Mexican tariff structure and without transferring a large economic burden to the lower income deciles.

While few, if any, countries have burdened their utilities with a tariff structure as complex and as sensible to climate as Mexico, our findings resonate for global electricity subsidy strategies. Any increasing block tariff scheme (the most common worldwide), where more consumption leads to more subsidies is vulnerable to overburdening and increasing regressivity due to larger thermal loads due to climate change. There is, however, a tradeoff where someone will pay for the climate burden, whether it be the utilities or the users, so tariff restructuring programs must take care not to strongly punish lower income deciles, taking in consideration climate change when designing any new tariff structure and subsidy program.

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Author contributions **Rodrigo Muñoz Sánchez**: Writing – original draft, software, conceptualization, methodology, formal analysis. **Julián A. Velasco**: Writing – review and editing, Supervision. **Francisco Estrada Porrúa**: Writing – review and editing, Supervision. **Oscar Calderón Bustamante**: Writing – review and editing, Supervision.

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Data availability Data will be made available on request. The code used for the data analysis is available at https://github.com/rodrigoms95/herramientas-climatico-22-1/tree/main/code/Proyecto/Jupyter.

Declarations

Competing interest The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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